Kernix 0.7.2 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

16 workloads

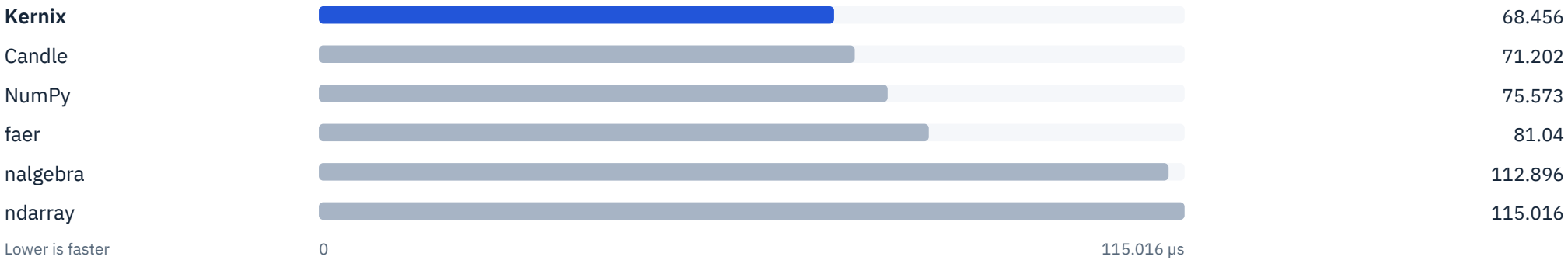
62 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

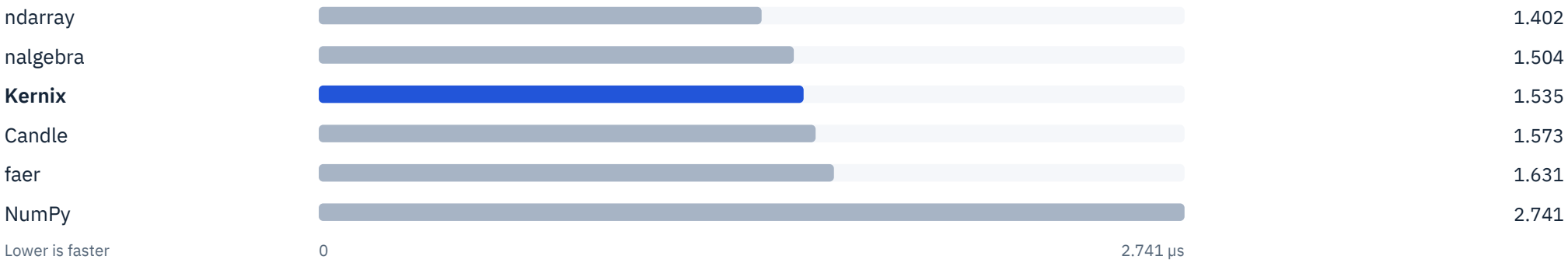
Kernix 0.7.2 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

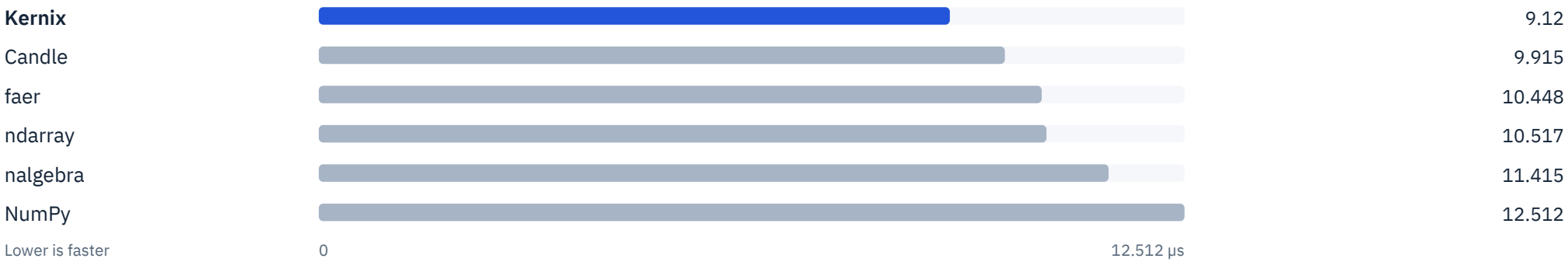
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



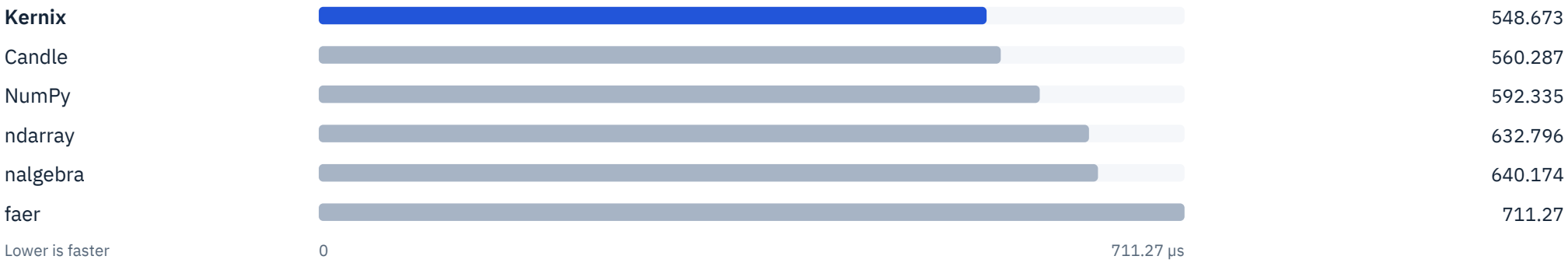
Kernix 0.7.2 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

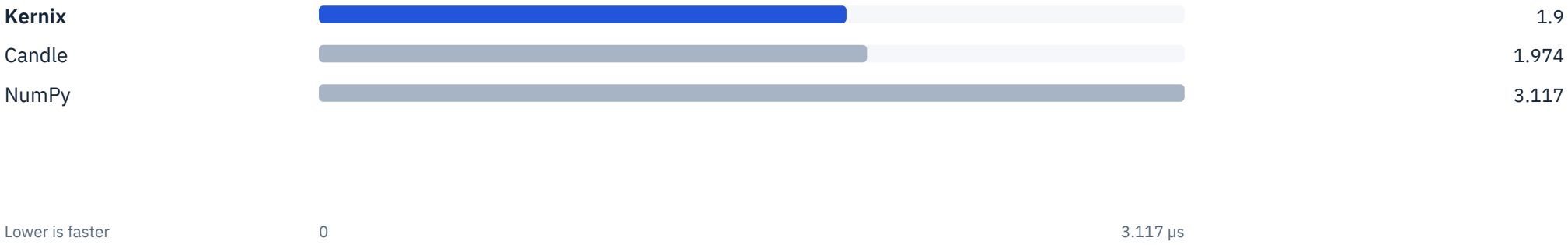
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

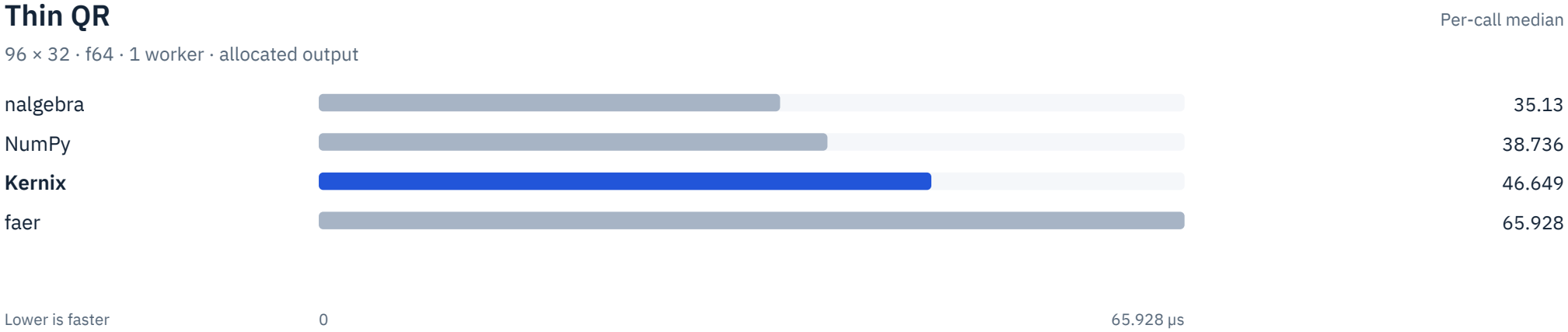


Kernix 0.7.2 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

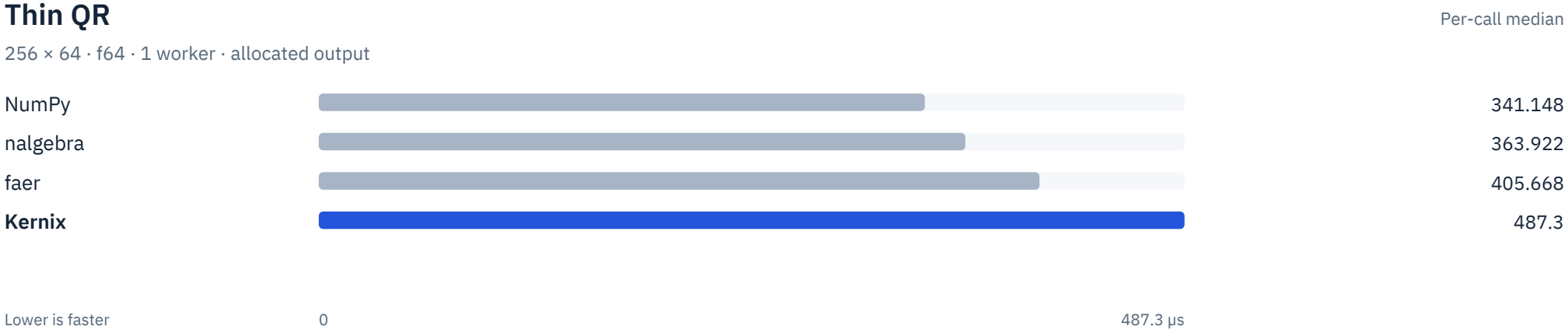
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output

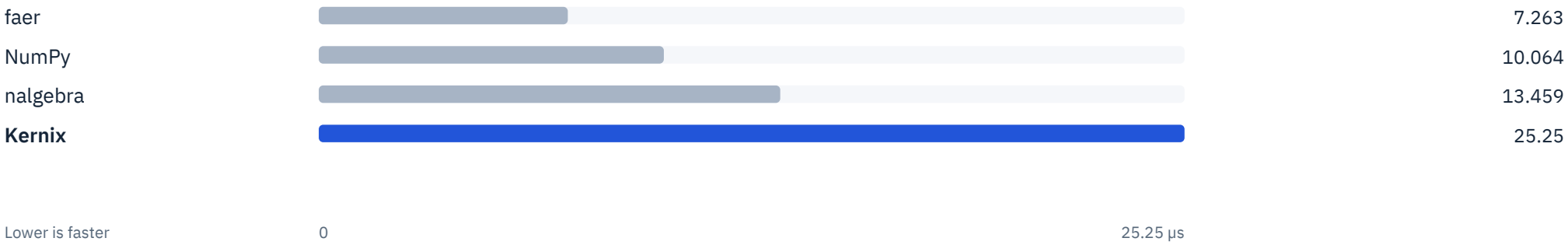


Kernix 0.7.2 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

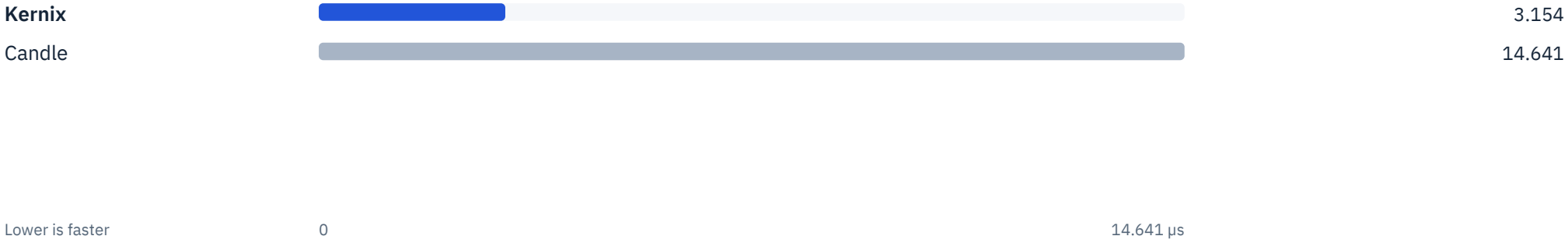
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.7.2 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

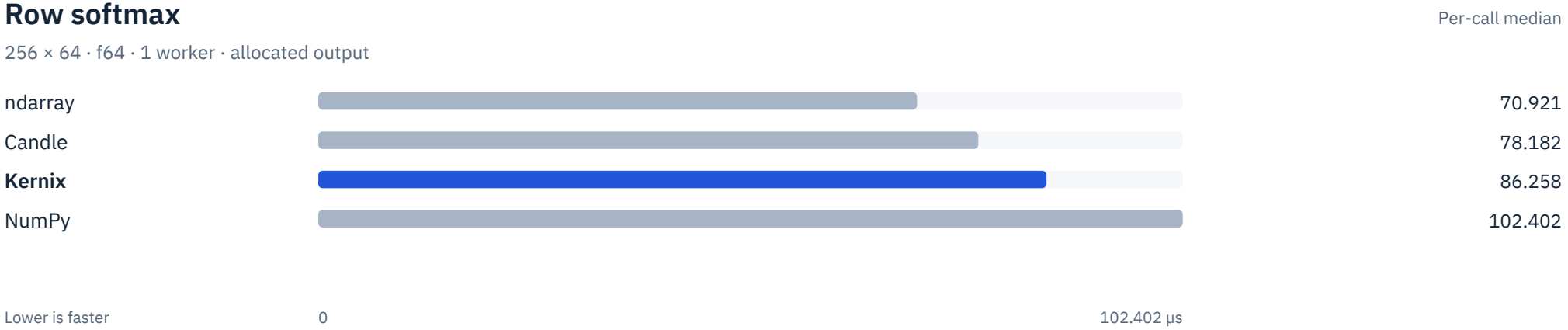


Kernix 0.7.2 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



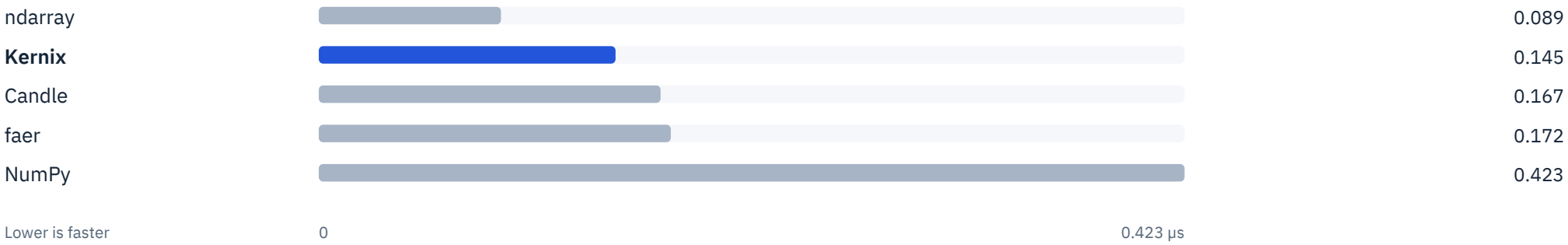
Kernix 0.7.2 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1535	1.535
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1402	1.402
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1504	1.504
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1631	1.631
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1573	1.573
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2741	2.741
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9120	9.12
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	10517	10.517
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	11415	11.415
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10448	10.448

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9915	9.915
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	12512	12.512
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	68456	68.456
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	115016	115.016
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	112896	112.896
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	81040	81.04
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	71202	71.202
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	75573	75.573
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	548673	548.673
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	632796	632.796

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	640174	640.174
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	711270	711.27
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	560287	560.287
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	592335	592.335
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1900	1.9
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	1974	1.974
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	3117	3.117
Thin QR / Kernix 96 × 32 · f64	46649	46.649
Thin QR / nalgebra 96 × 32 · f64	35130	35.13
Thin QR / faer 96 × 32 · f64	65928	65.928

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 96 × 32 · f64	38736	38.736
Thin QR / Kernix 256 × 64 · f64	487300	487.3
Thin QR / nalgebra 256 × 64 · f64	363922	363.922
Thin QR / faer 256 × 64 · f64	405668	405.668
Thin QR / NumPy 256 × 64 · f64	341148	341.148
Cholesky factorization / Kernix 64 × 64 · f64	25250	25.25
Cholesky factorization / nalgebra 64 × 64 · f64	13459	13.459
Cholesky factorization / faer 64 × 64 · f64	7263	7.263
Cholesky factorization / NumPy 64 × 64 · f64	10064	10.064
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2382	2.382

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	6678	6.678
Model inference / Kernix batched normalized/sliced MLP · f64	1966	1.966
Model inference / Candle batched normalized/sliced MLP · f64	3479	3.479
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2438	2.438
Model recording + forward / Candle batched normalized/sliced MLP · f64	5299	5.299
Model backward / Kernix prepared graph → fresh gradients · f64	3154	3.154
Model backward / Candle prepared graph → fresh gradients · f64	14641	14.641
Parameter update / Kernix transactional SGD, fixed gradients · f64	489	0.489
Row softmax / Kernix 256 × 64 · f64	86258	86.258
Row softmax / ndarray 256 × 64 · f64	70921	70.921

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / Candle 256 × 64 · f64	78182	78.182
Row softmax / NumPy 256 × 64 · f64	102402	102.402
Create zeros / Kernix 64 × 64 · f32	145	0.145
Create zeros / ndarray 64 × 64 · f32	89	0.089
Create zeros / faer 64 × 64 · f32	172	0.172
Create zeros / Candle 64 × 64 · f32	167	0.167
Create zeros / NumPy 64 × 64 · f32	423	0.423
Create ones / Kernix 64 × 64 · f32	185	0.185
Create ones / ndarray 64 × 64 · f32	144	0.144
Create ones / faer 64 × 64 · f32	201	0.201

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / Candle 64 × 64 · f32	237	0.237
Create ones / NumPy 64 × 64 · f32	1234	1.234